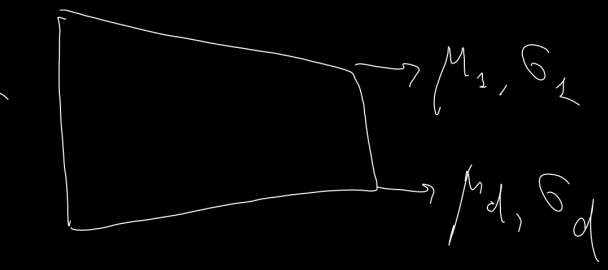
Policy Gradient Methods

DQN: $\operatorname{argmax}_a Q(s,a)$

PG = learn policy $\pi(a|s,\theta)$

$a \in \mathbb{R}^d$



$$\pi(a|s,\theta) = \prod_{j=1}^{d} N(a_j | \mu_j, \text{softplus}(\sigma_j))$$

$a \in \{1, 2, \ldots, A\}$ $\quad \pi(a|s,\theta) = \text{Softmax}(\text{outputs}(s,\theta))$

REINFORCE

$$J(\theta) = \mathbb{E}_{p(\tau|\theta)} R(\tau) \to \max_{\theta}$$

$$R(\tau) = r_0 + \gamma r_1 + \dots + \gamma^T r_T$$

$$p(\tau|\theta) = p(s_0) \prod_{t=0}^{T-1} \pi(a_t|s_t,\theta)\, p(s_{t+1}|s_t, a_t)$$

$$\nabla_\theta J(\theta) = \nabla_\theta \mathbb{E}_{p(\tau|\theta)} R(\tau) = \left\{ \underset{\text{trick}}{\text{log-derivative}} \right\} =$$

$$= \nabla_\theta \int p(\tau|\theta)\, R(\tau)\, d\tau =$$

$$= \int \nabla_\theta p(\tau|\theta) R(\tau) d\tau = \int p(\tau|\theta) \frac{\nabla_\theta p(\tau|\theta)}{p(\tau|\theta)} R(\tau) d\tau =$$

$$= \mathbb{E}_{p(\tau|\theta)} \underbrace{\nabla_\theta \log p(\tau|\theta)}\, R(\tau) =$$

$$= \mathbb{E}_{p(\tau|\theta)} \left[\sum_{t=0}^{T} \nabla_\theta \log \pi(a_t|s_t,\theta) \right] \left[\sum_{s=0}^{T} \gamma^s r_s \right] =$$

$$= \sum_{t,s} \mathbb{E}_{p(\tau|\theta)} \nabla \log \pi(a_t|s_t,\theta)\, \gamma^s r_s \quad (=)$$

$t > s$

$$\mathbb{E}_{s_0, a_0, s_1, a_1, \ldots, s_t} \gamma^s r_s \underbrace{\mathbb{E}_{a_t} \nabla \log \pi(a_t|s_t,\theta)}$$

$$\mathbb{E}_{a_t} \nabla \log \pi(a_t|s_t,\theta) = \sum_{a_t} \pi(a_t|s_t,\theta) \frac{\nabla_\theta \pi(a_t|s_t,\theta)}{\pi(a_t|s_t,\theta)} = \nabla_\theta \underbrace{\sum_{a_t} \pi(a_t|s_t,\theta)}_{1} = 0$$

$$= \sum_t \mathbb{E}_{p(\tau|\theta)} \nabla \log \pi(a_t|s_t,\theta) \underbrace{\left(\sum_{s=t}^{T} \gamma^s r_s \right)}_{\gamma^t \sum_{s=t}^{T} \gamma^{s-t} r_s}$$

REINFORCE scheme

Init. θ

For episodes:

  Init. s_0

  For $t = 0, 1, \ldots, T$:

    make $a_t \sim \pi(a_t|s_t,\theta)$, get r_t, s_{t+1}

  $R := 0, L = 0$

  For $t = T-1, T-2, \ldots, 0$:

    $R = r_t + \gamma R$

    $L = L + \gamma^t R \log \pi(a_t|s_t,\theta)$

$\theta = \text{opt_step}(\nabla_\theta L)$

Example

$\mathbb{E}_{\pi(a)} f_1(a) \cdot f_2(a)$

a	0	1
$\pi(a)$	$\frac{1}{2}$	$\frac{1}{2}$

$$f_1 = \begin{cases} -1, & \text{if } a=0 \\ +1, & \text{if } a=1 \end{cases}$$

$$f_2 = \begin{cases} 100, & \text{if } a=0 \\ 101, & \text{if } a=1 \end{cases}$$

$$\mathbb{E}_{\pi(a)} f_1(a) = \frac{1}{2}(-1+1) = 0$$

$$\mathbb{E}_{\pi(a)} f_1(a) f_2(a) = \frac{1}{2}\left((-1)\cdot 100 + 1\cdot 101\right) =$$

$$= \frac{1}{2}$$

$$Var_2 = \mathbb{E}_{\pi(a)}\left(f_1(a) f_2(a) - \frac{1}{2}\right)^2 =$$

$$= \frac{1}{2}\left((-1)\cdot 100 - \frac{1}{2}\right)^2 + \frac{1}{2}\left(+1\cdot 101 - \frac{1}{2}\right)^2 = 100.5^2$$

$$\mathbb{E}_{\pi(a)} f_1(a)\left(f_2(a) - \mathbb{E}_{\pi(a)} f_2(a)\right) = \mathbb{E}_{\pi(a)} f_1(a)\cdot f_2(a)$$

$$\mathbb{E}_{\pi(a)} f_2(a) = \frac{1}{2}(100+101) = 100.5$$

$$\mathrm{Var} = \mathbb{E}_{\pi(a)}\left(f_1(a)(f_2(a)-100.5) - \frac{1}{2}\right)^2 =$$

$$= \frac{1}{2}\underbrace{\left((-1)(100-100.5) - \frac{1}{2}\right)^2}_{0} + \frac{1}{2}\underbrace{\left((+1)(101-100.5) - \frac{1}{2}\right)^2}_{0} = 0$$

PG main theorem

$$J(\theta) = \mathbb{E}_{p(s_0)} \mathbb{E}_{\pi(a_0|s_0,\theta)} Q^{\pi}(s_0, a_0)$$

$$\nabla_\theta J(\theta) = \nabla_\theta \mathbb{E}_{p(s_0)} \sum_{a_0} \pi(a_0|s_0,\theta) Q^{\pi}(s_0, a_0) =$$

$$= \mathbb{E}_{p(s_0)} \sum_{a_0} \left[\nabla_\theta \pi(a_0|s_0,\theta) \cdot Q^{\pi}(s_0, a_0) + \pi(a_0|s_0,\theta) \nabla_\theta Q^{\pi}(s_0, a_0) \right] =$$

$$= \mathbb{E}_{p(s_0)} \sum_{a_0} \left[\nabla \pi(a_0|s_0,\theta) Q^{\pi}(s_0, a_0) + \pi(a_0|s_0,\theta) \nabla_\theta \left(r(s_0, a_0) + \gamma \mathbb{E}_{p(s_1|s_0,a_0)} \mathbb{E}_{\pi(a_1|s_1,\theta)} Q^{\pi}(s_1, a_1) \right) \right] =$$

$$= \mathbb{E}_{p(s_0)} \sum_{a_0} \Big[\nabla \pi(a_0|s_0,\theta)\, Q^{\pi}(s_0,a_0) + \pi(a_0|s_0,\theta)\, \gamma\, \mathbb{E}_{p(s_1|s_0,a_0)} \nabla_{\theta} \mathbb{E}_{\pi(a_1|s_1,\theta)} Q^{\pi}(s_1,a_1) \Big] =$$

$$= \ldots = \sum_{s} \Big(\sum_{k=0}^{+\infty} \gamma^k p(s_0 \to s, k, \pi) \Big) \sum_{a} \nabla \pi(a|s,\theta)\, Q^{\pi}(s,a) \quad (*)$$

$\tilde{\mu}^{\pi}(s)$ — prob. to come to state s in k steps under policy π

$$\sum_{s} \tilde{\mu}^{\pi}(s) = \sum_{s} \sum_{k} \gamma^k p(s_0 \to s, k, \pi) = \sum_{k} \gamma^k \underbrace{\sum_{s} p(s_0 \to s, k, \pi)}_{1} = \frac{1}{1-\gamma}$$

$\mu^{\pi}(s) := (1-\gamma)\, \tilde{\mu}^{\pi}(s)$ — discounted state visitation distr.

$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim \mu^{\pi}(s)} \sum_{a} \nabla_{\theta} \pi(a|s,\theta) \cdot Q^{\pi}(s,a) =$$

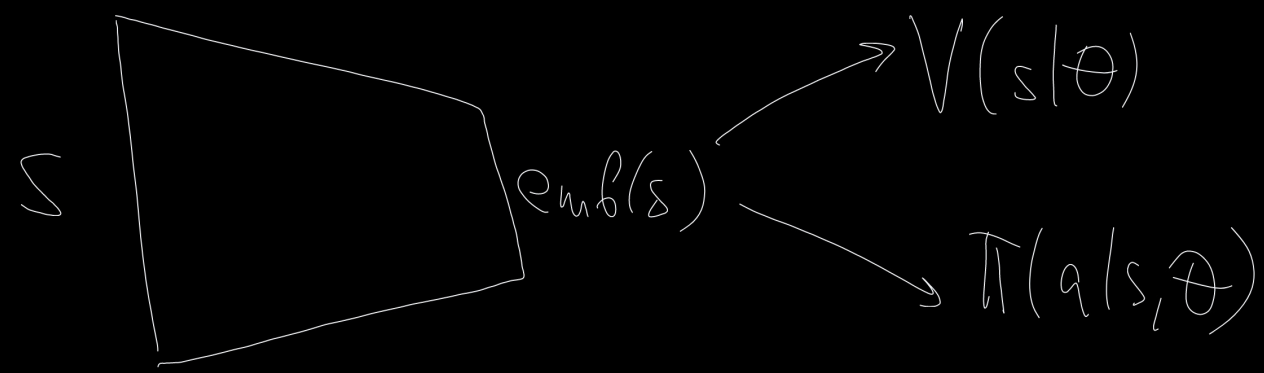
$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim \mu^{\pi}(s)} \mathbb{E}_{a \sim \pi(a|s,\theta)} \nabla_{\theta} \log \pi(a|s,\theta) \cdot Q^{\pi}(s,a)$$

A2C (Advantage Actor-Critic)

$$\mathbb{E}_{s\sim\mu^{\pi}(s)}\,\mathbb{E}_{a\sim\pi(a|s,\theta)}\,\nabla_{\theta}\log\pi(a|s,\theta)\left(Q^{\pi}(s,a)-V^{\pi}(s)\right)$$

$$=\mathbb{E}_{a\sim\pi(a|s,\theta)}\,Q^{\pi}(s,a)$$

Advantage $A^{\pi}(s,a) := Q^{\pi}(s,a) - V^{\pi}(s)$



$$GAE_{t:t+n} = r_t + \gamma r_{t+1} + \dots + \gamma^{n-1} r_{t+n-1} + \gamma^n V(s_{t+n}) - V(s_t)$$

$$GAE_t^\lambda = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} GAE_{t:t+n}$$

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

$$GAE_t^\lambda = \sum_{k=t}^{\infty} (\gamma\lambda)^{k-t} \delta_k$$

A2C scheme

$n = 5$

$\#\,envs = 8$

Init. θ

For iterations:

$B := \emptyset$

For each parallel rollout:

Collect $s_0, r_0, a_0, \ldots, s_{n-1}, r_{n-1}, a_{n-1}, s_n$ using $\pi(a|s, \theta)$

$GAE_n := 0$

For $t = n-1, n-2, \ldots, 0$:

$\delta_t = r_t + V(s_{t+1}|\theta) - V(s_t|\theta)$

$GAE_t = \delta_t + (\gamma\lambda) \cdot GAE_{t+1}$

$y_t = GAE_t + V(s_t|\theta)$

Add $\{s_t, a_t, GAE_t, y_t\}$ to batch B

$$GAE = Normalize(GAE)$$

$$c_1 = \frac{1}{2}$$

$$c_2 = 0.01$$

for $t \in B$:

$$L_\theta^{policy} = L_\theta^{policy} - \log \pi(a_t | s_t, \theta) \cdot GAE_t$$

$$L_\theta^{value} = L_\theta^{value} + \left(y_t - V(s_t | \theta)\right)^2$$

$$Entropy = Entropy - \sum_{a_t} \pi(a_t | s_t, \theta) \log \pi(a_t | s_t, \theta)$$

$$L_\theta = L_\theta^{policy} + c_1 L_\theta^{value} - c_2 Entropy$$

$$\theta = opt_step(\nabla_\theta L)$$